



1. Writing Component

1.1 Search with Heuristics

a. $\#state = \binom{n^2}{n} \approx 2^{O(n \log n)}$.

b. We have $h_i(x_i, y_i) = \text{dis}((x_i, y_i), (n - i + 1, n))$, so it is admissible. When there is other cars, it is not admissible since jump over a car can have a smaller cost.

c.

i. $\sum h_i$ is not admissible. Consider $(n - 1)$ -th car is at $(2, n - 2)$ and n -th car is at $(2, n - 1)$, other cars is arrived, then it costs 3 but the formular gets 4.

ii. $\min h_i$ is admissible. If there is only 1 car not arrived, then $\min h_i$ is actually the cost. If there is more than 2 car not arrived, then the cost is at least $\frac{1}{2} \sum h_i$, and $\min h_i \leq \frac{1}{2} \sum h_i$.

iii. $\max h_i$ is not admissible. Consider $n - 2$ -th car is at $(3, n - 1)$ and n -th car is at $(3, n - 2)$, other cars is arrived, then it costs 3 but the formular gets 4.

iv. $n \min h_i$ is not admissible. The same counter example as i.

v. $n \max h_i$ is not admissible. The same counter example as iv.

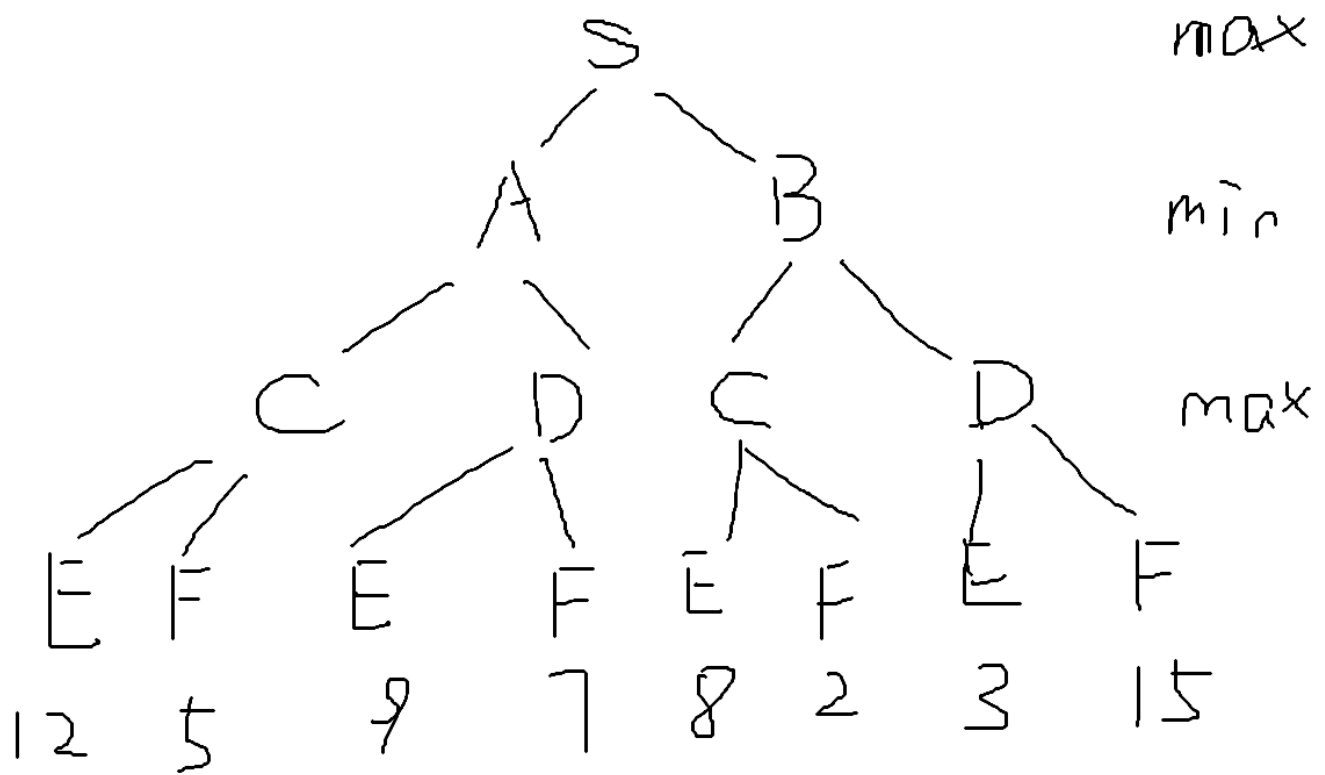
1.2 Adjusted Heuristics

The example on the slide is enough. Consider $(S, A, 1), (S, B, 1), (A, C, 1), (B, C, 2), (C, G, 3)$ with $h_S = 2, h_A = 4, h_B = 1, h_C = 1$.

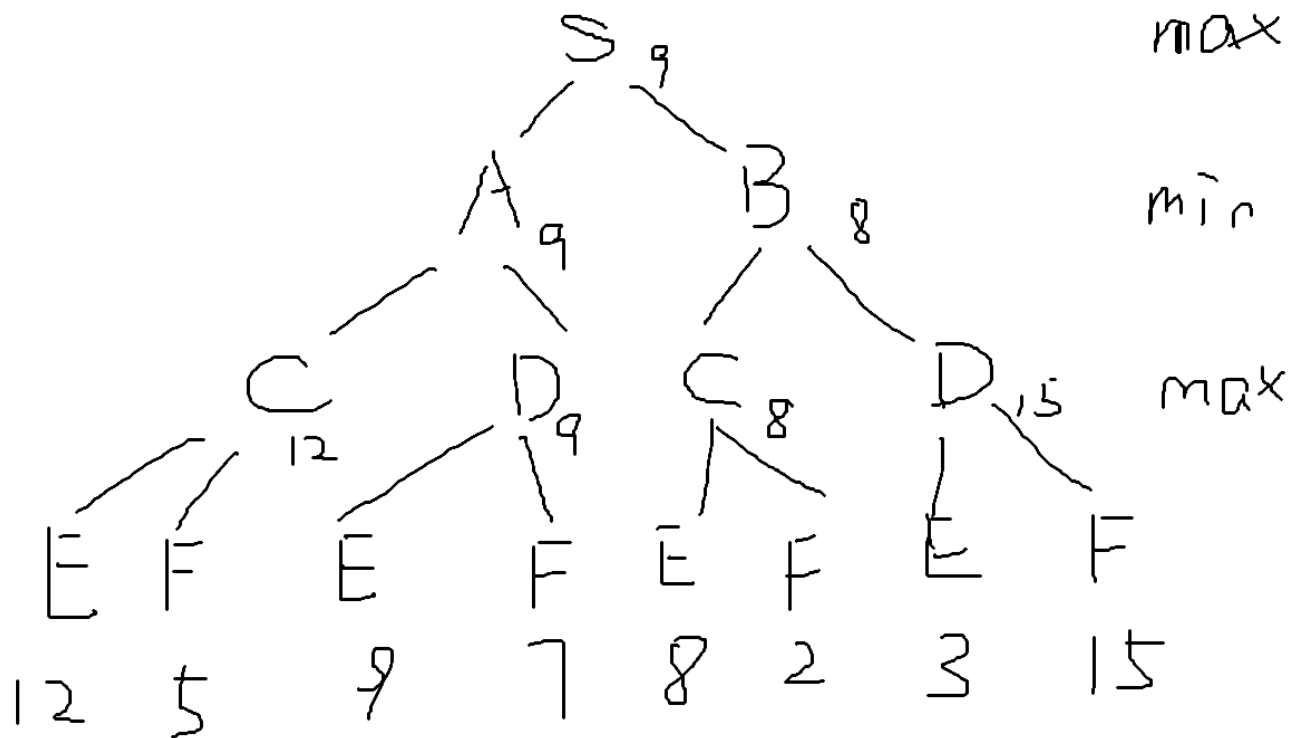
The first time we expand C is by $S - B - C$ and is longer than $S - A - C$. Although we will increase h_C , but it's too late.

1.3 Adversarial Search

a.

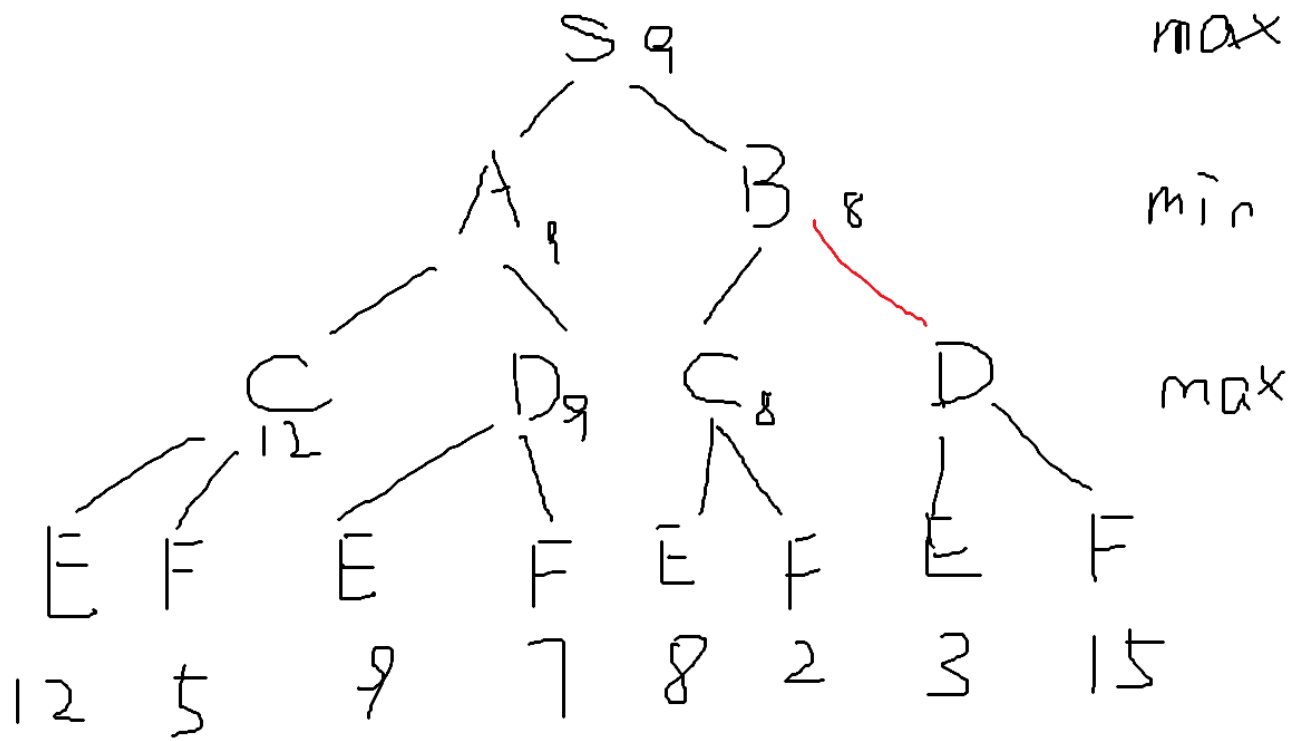


b.



So the root is 9.

c.



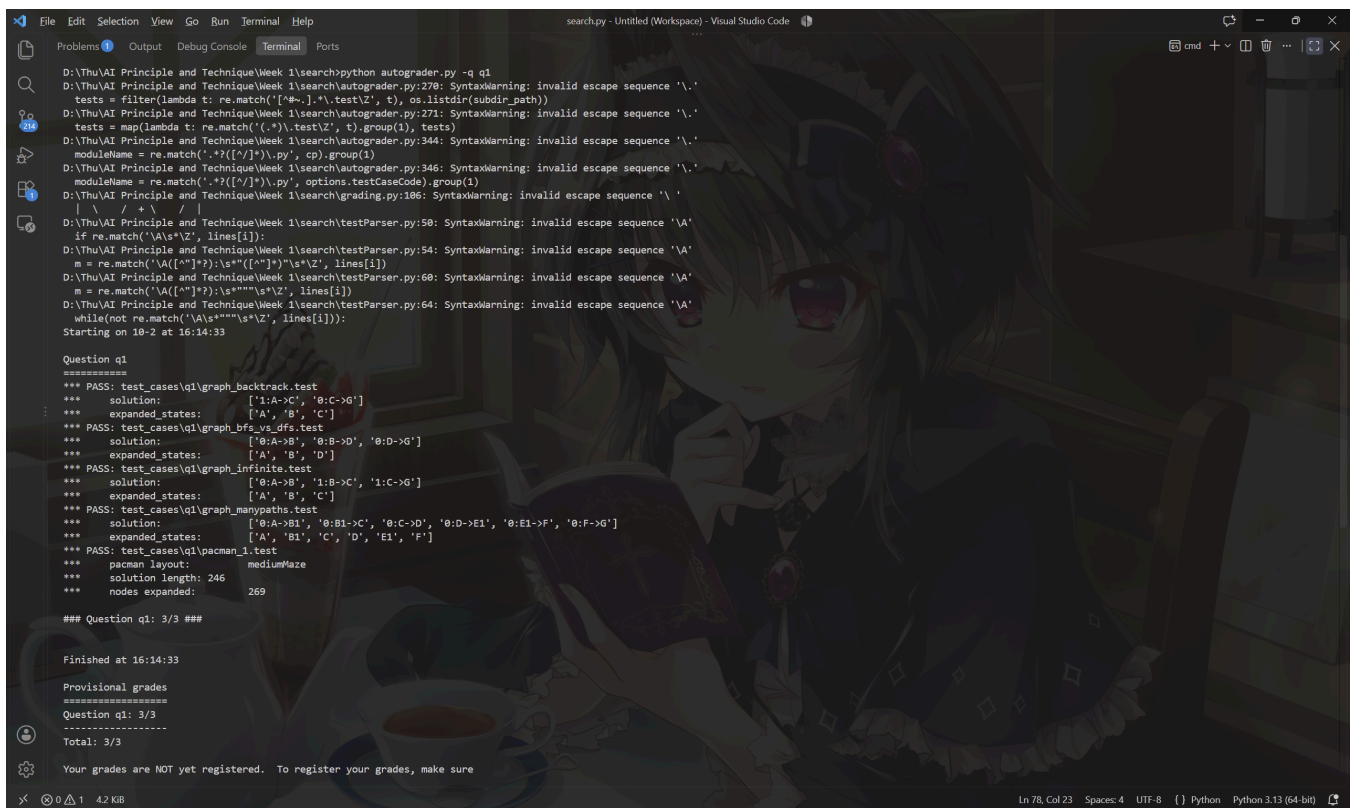
The red edge is pruned.

2. Programming Component 1: Search in Pacman

2.1 Introduction

2.2 Welcome to Pacman

2.3 Question 1: Finding a Fixed Food Dot using Depth First Search



```
File Edit Selection View Go Run Terminal Help
search.py - Untitled (Workspace) - Visual Studio Code

D:\ThuVAI Principle and Technique\Week 1\search\python autograder.py -q q1
D:\ThuVAI Principle and Technique\Week 1\search\autograder.py:278: SyntaxWarning: invalid escape sequence '\.'
  tests = filter(lambda t: re.match('[^*].*\.\test(Z', t), os.listdir(subdir_path))
D:\ThuVAI Principle and Technique\Week 1\search\autograder.py:271: SyntaxWarning: invalid escape sequence '\.'
  tests = map(lambda t: re.match('(.*)\test(Z', t).group(1), tests)
D:\ThuVAI Principle and Technique\Week 1\search\autograder.py:344: SyntaxWarning: invalid escape sequence '\.'
  moduleName = re.match('.*?([/\\]*).py', cp).group(1)
D:\ThuVAI Principle and Technique\Week 1\search\autograder.py:346: SyntaxWarning: invalid escape sequence '\.'
  moduleName = re.match('.*?([/\\]*).py', options.testCaseCode).group(1)
D:\ThuVAI Principle and Technique\Week 1\search\grading.py:106: SyntaxWarning: invalid escape sequence '\ '
  | \ / + \ / |
D:\ThuVAI Principle and Technique\Week 1\search\testParser.py:50: SyntaxWarning: invalid escape sequence '\A'
  if re.match('\As+Z', lines[i]):
D:\ThuVAI Principle and Technique\Week 1\search\testParser.py:54: SyntaxWarning: invalid escape sequence '\A'
  m = re.match('\A([*]*)?:\s*""([*]*)"\s+Z', lines[i])
D:\ThuVAI Principle and Technique\Week 1\search\testParser.py:60: SyntaxWarning: invalid escape sequence '\A'
  m = re.match('\A([*]*)?:\s*""\s+Z', lines[i])
D:\ThuVAI Principle and Technique\Week 1\search\testParser.py:64: SyntaxWarning: invalid escape sequence '\A'
  while(not re.match('\As+""\s+Z', lines[i])):
Starting on 10-2 at 16:14:33

Question q1
=====
*** PASS: test_cases\q1\graph_backtrack.test
*** solution: ['1:A->C', '0:C->G']
*** expanded_states: ['A', 'B', 'C']
*** PASS: test_cases\q1\graph_bfs_vs_dfs.test
*** solution: ['0:A->B', '0:B->D', '0:D->G']
*** expanded_states: ['A', 'B', 'D']
*** PASS: test_cases\q1\graph_infinite.test
*** solution: ['0:A->B', '1:B->C', '1:C->G']
*** expanded_states: ['A', 'B', 'C']
*** PASS: test_cases\q1\graph_manypaths.test
*** solution: ['0:A->B1', '0:B1->C', '0:C->D', '0:D->E1', '0:E1->F', '0:F->G']
*** expanded_states: ['A', 'B1', 'C', 'D', 'E1', 'F']
*** PASS: test_cases\q1\pacman_1.test
*** pacman layout: mediumMaze
*** solution length: 246
*** nodes expanded: 269

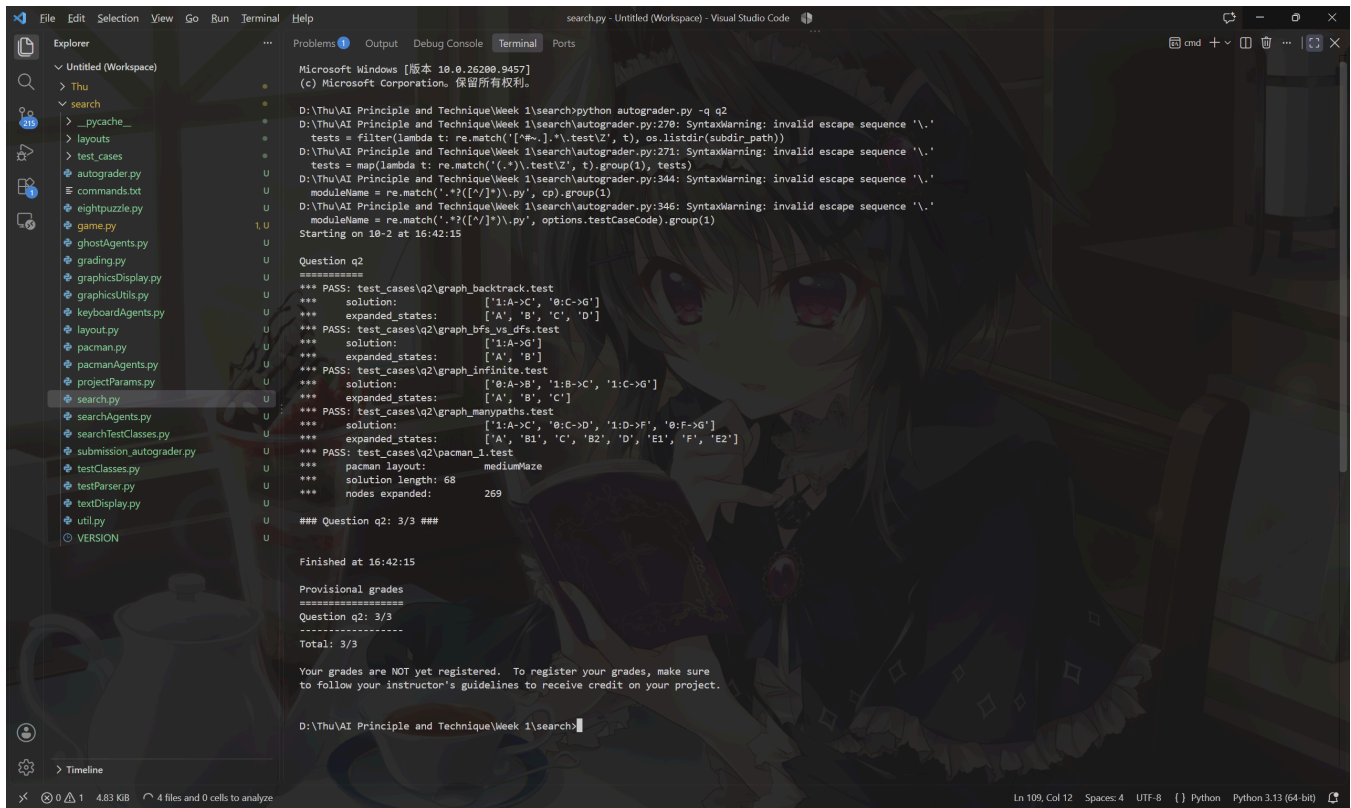
### Question q1: 3/3 ###

Finished at 16:14:33

Provisional grades
=====
Question q1: 3/3
-----
Total: 3/3

Your grades are NOT yet registered. To register your grades, make sure
```

2.4 Question 2: Breadth-First Search



```
search.py - Untitled (Workspace) - Visual Studio Code
Microsoft Windows [版本 10.0.26290.9457]
(c) Microsoft Corporation. 保留所有权利。

D:\ThuVAI Principle and Technique\Week 1\search>python autograder.py -q q2
D:\ThuVAI Principle and Technique\Week 1\search\autograder.py:270: SyntaxWarning: invalid escape sequence '\.'
  tests = filter(lambda t: re.match('[^#-].*\.\test[V]', t), os.listdir(subdir_path))
D:\ThuVAI Principle and Technique\Week 1\search\autograder.py:271: SyntaxWarning: invalid escape sequence '\.'
  tests = map(lambda t: re.match('(.*)\.\test[V]', t).group(1), tests)
D:\ThuVAI Principle and Technique\Week 1\search\autograder.py:344: SyntaxWarning: invalid escape sequence '\.'
  moduleName = re.match('.*?([?/])\.\py', cp).group(1)
D:\ThuVAI Principle and Technique\Week 1\search\autograder.py:346: SyntaxWarning: invalid escape sequence '\.'
  moduleName = re.match('.*?([?/])\.\py', options.testCaseCode).group(1)
Starting on 10-2 at 16:42:15

Question q2
=====
*** PASS: test_cases\q2\graph_backtrack.test
*** solution:      ['1:A->C', '0:C->G']
*** expanded_states: ['A', 'B', 'C', 'D']
*** PASS: test_cases\q2\graph_bfs_vs_dfs.test
*** solution:      ['1:A->G']
*** expanded_states: ['A', 'B']
*** PASS: test_cases\q2\graph_infinite.test
*** solution:      ['0:A->B', '1:B->C', '1:C->G']
*** expanded_states: ['A', 'B', 'C']
*** PASS: test_cases\q2\graph_manypaths.test
*** solution:      ['1:A->C', '0:C->D', '1:D->F', '0:F->G']
*** expanded_states: ['A', 'B1', 'C', 'B2', 'D', 'E1', 'F', 'E2']
*** PASS: test_cases\q2\pacman_1.test
*** pacman layout:      mediumMaze
*** solution length: 68
*** nodes expanded:      269

### Question q2: 3/3 ###

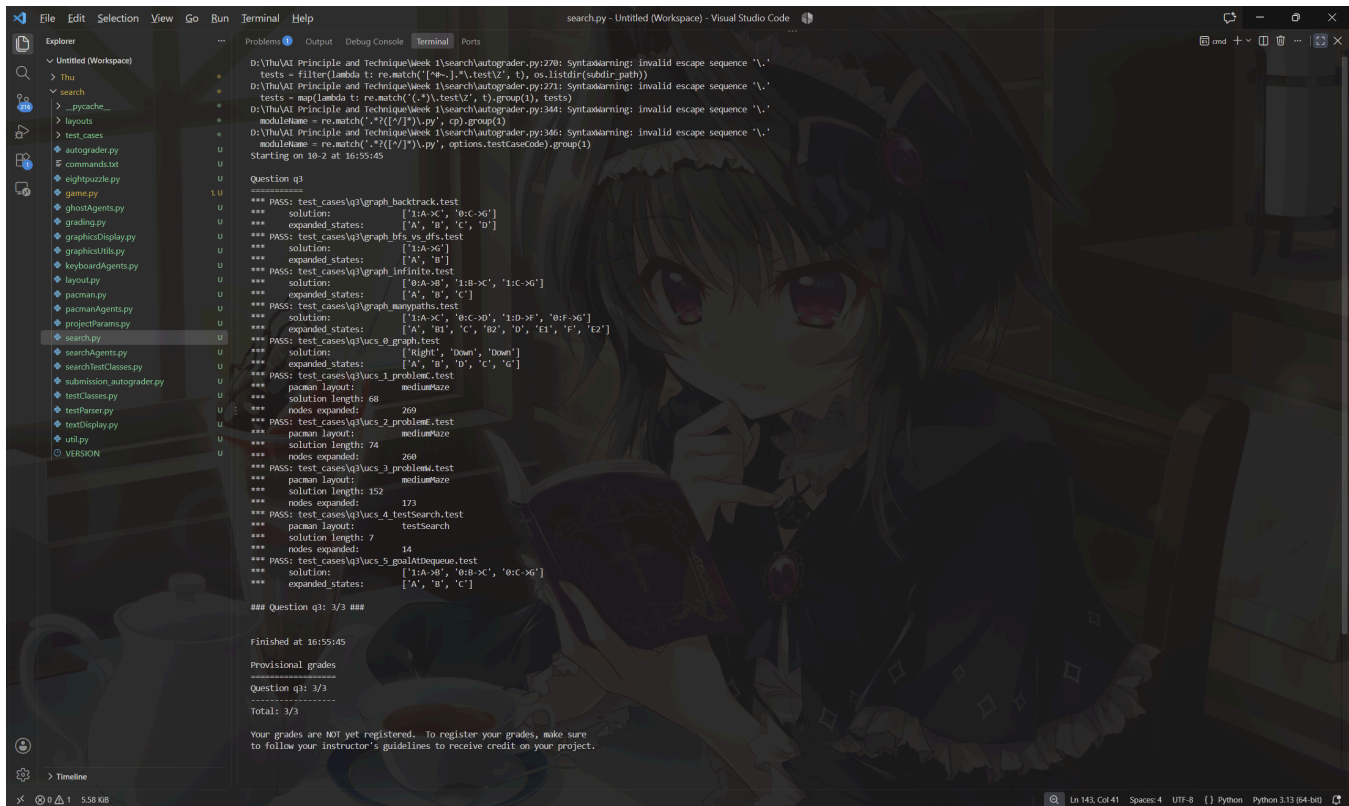
Finished at 16:42:15

Provisional grades
=====
Question q2: 3/3
-----
Total: 3/3

Your grades are NOT yet registered. To register your grades, make sure
to follow your instructor's guidelines to receive credit on your project.

D:\ThuVAI Principle and Technique\Week 1\search>
```

2.5 Question 3: Varying the Cost Function



```
search.py - Untitled (Workspace) - Visual Studio Code

File Edit Selection View Go Run Terminal Help

Explorer
  Untitled (Workspace)
  Thru
  search
  _pycache_
  layouts
  test_cases
  autograder.py
  command.py
  nighpuzzle.py
  pacman.py
  ghostAgents.py
  grading.py
  graphicsDisplay.py
  graphicsUtils.py
  keyboardAgents.py
  layout.py
  pacman.py
  pacmanAgents.py
  projectParams.py
  search.py
  searchAgents.py
  searchTestClasses.py
  submission_autograder.py
  testClasses.py
  testParser.py
  testDisplay.py
  util.py
  VERSION

Terminal
  D:\VHUAI\Principle and Technique\Week 1\search\autograder.py:270: SyntaxWarning: invalid escape sequence '\.'
  tests = filter(lambda t: re.match('[*~.-]*\.', test\2', t), os.listdir(subdir_path))
  D:\VHUAI\Principle and Technique\Week 1\search\autograder.py:271: SyntaxWarning: invalid escape sequence '\.'
  tests = map(lambda t: re.match('[*~.-]*\.', test\2', t).group(1), tests)
  D:\VHUAI\Principle and Technique\Week 1\search\autograder.py:344: SyntaxWarning: invalid escape sequence '\.'
  modulelame = re.match('[*~.-]*\.', cp).group(1)
  D:\VHUAI\Principle and Technique\Week 1\search\autograder.py:345: SyntaxWarning: invalid escape sequence '\.'
  modulelame = re.match('[*~.-]*\.', cp).group(1)
  Starting on 10-2 at 16:55:45

  Question q1
  =====
  *** PASS: test_cases\q1\graph_backtrack.test
  *** solution: ['1A-3C', '01C-3G']
  *** expanded_states: ['A', 'B', 'C', 'D']
  *** PASS: test_cases\q1\graph_bfs_vs_dfs.test
  *** solution: ['1A-3G']
  *** expanded_states: ['A', 'B']
  *** PASS: test_cases\q1\graph_infinite.test
  *** solution: ['01A-3B', '11B-3C', '11C-3G']
  *** expanded_states: ['A', 'B', 'C']
  *** PASS: test_cases\q1\graph_manipaths.test
  *** solution: ['11A-3C', '01C-3D', '11D-3E', '01F-3G']
  *** expanded_states: ['A', 'B', 'C', 'D', 'E', 'F', 'G']
  *** PASS: test_cases\q1\ucs_0_graph.test
  *** solution: ['right', 'down', 'down']
  *** expanded_states: ['A', 'B', 'D', 'C', 'G']
  *** PASS: test_cases\q1\ucs_1_problem.test
  *** pacman layout: mediummaze
  *** solution length: 68
  *** nodes expanded: 269
  *** PASS: test_cases\q1\ucs_2_problem.test
  *** solution length: 74
  *** nodes expanded: 260
  *** PASS: test_cases\q1\ucs_3_problem.test
  *** pacman layout: mediummaze
  *** solution length: 152
  *** nodes expanded: 173
  *** PASS: test_cases\q1\ucs_4_testSearch.test
  *** pacman layout: testSearch
  *** solution length: 7
  *** nodes expanded: 14
  *** PASS: test_cases\q1\ucs_5_goalAtDeque.test
  *** solution: ['11A-3B', '01B-3C', '01C-3G']
  *** expanded_states: ['A', 'B', 'C']

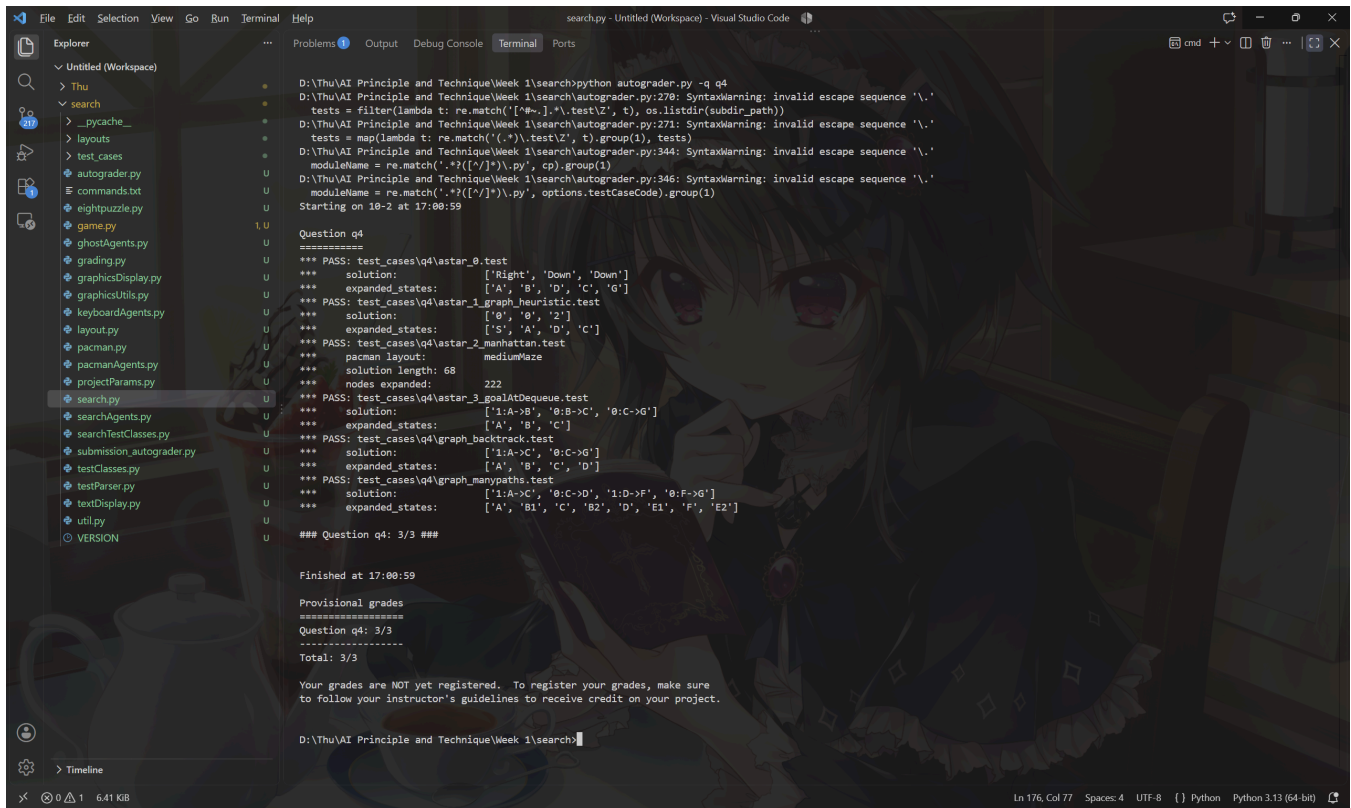
  ## Question q1: 3/3 ##

  Finished at 16:55:45

  Provisional grades
  =====
  Question q1: 3/3
  Total: 3/3

  Your grades are NOT yet registered.. To register your grades, make sure
  to follow your instructor's guidelines to receive credit on your project.
```

2.6 Question 4: A* search



```
search.py - Untitled (Workspace) - Visual Studio Code
File Edit Selection View Go Run Terminal Help
Explorer
  Untitled (Workspace)
  > Thu
  > search
  > __pycache__
  > layouts
  > test_cases
  > autograder.py
  > commands.txt
  > eightpuzzle.py
  > game.py
  > ghostAgents.py
  > grading.py
  > graphicsDisplay.py
  > graphicsUtils.py
  > keyboardAgents.py
  > layout.py
  > pacman.py
  > pacmanAgents.py
  > projectParams.py
  > search.py
  > searchAgents.py
  > searchTestClasses.py
  > submission_autograder.py
  > testClasses.py
  > testParser.py
  > textDisplay.py
  > util.py
  > VERSION

Terminal
D:\Thu\AI Principle and Technique\Week 1\search>python autograder.py -q q4
D:\Thu\AI Principle and Technique\Week 1\search\autograder.py:270: SyntaxWarning: invalid escape sequence '\.'
  tests = filter(lambda t: re.match('[^#-].*\test[Z]', t), os.listdir(subdir_path))
D:\Thu\AI Principle and Technique\Week 1\search\autograder.py:271: SyntaxWarning: invalid escape sequence '\.'
  tests = map(lambda t: re.match('(.*?)\test[Z]', t).group(1), tests)
D:\Thu\AI Principle and Technique\Week 1\search\autograder.py:344: SyntaxWarning: invalid escape sequence '\.'
  moduleName = re.match('.*?([/]*).*\.py', cp).group(1)
D:\Thu\AI Principle and Technique\Week 1\search\autograder.py:346: SyntaxWarning: invalid escape sequence '\.'
  moduleName = re.match('.*?([/]*).*\.py', options.testCaseCode).group(1)
Starting on 10-2 at 17:00:59

Question q4
=====
*** PASS: test_cases\q4\astar_0.test
*** solution:      ['Right', 'Down', 'Down']
*** expanded_states: ['A', 'B', 'D', 'C', 'G']
*** PASS: test_cases\q4\astar_1_graph_heuristic.test
*** solution:      ['0', '0', '2']
*** expanded_states: ['S', 'A', 'D', 'C']
*** PASS: test_cases\q4\astar_2_manhattan.test
*** pacman layout:      mediumMaze
*** solution length: 68
*** nodes expanded:      222
*** PASS: test_cases\q4\astar_3_goalAtDequeue.test
*** solution:      ['1:A->B', '0:B->C', '0:C->G']
*** expanded_states: ['A', 'B', 'C']
*** PASS: test_cases\q4\graph_backtrack.test
*** solution:      ['1:A->C', '0:C->G']
*** expanded_states: ['A', 'B', 'C', 'D']
*** PASS: test_cases\q4\graph_manypaths.test
*** solution:      ['1:A->C', '0:C->D', '1:D->F', '0:F->G']
*** expanded_states: ['A', 'B1', 'C', 'B2', 'D', 'E1', 'F', 'E2']

### Question q4: 3/3 ###

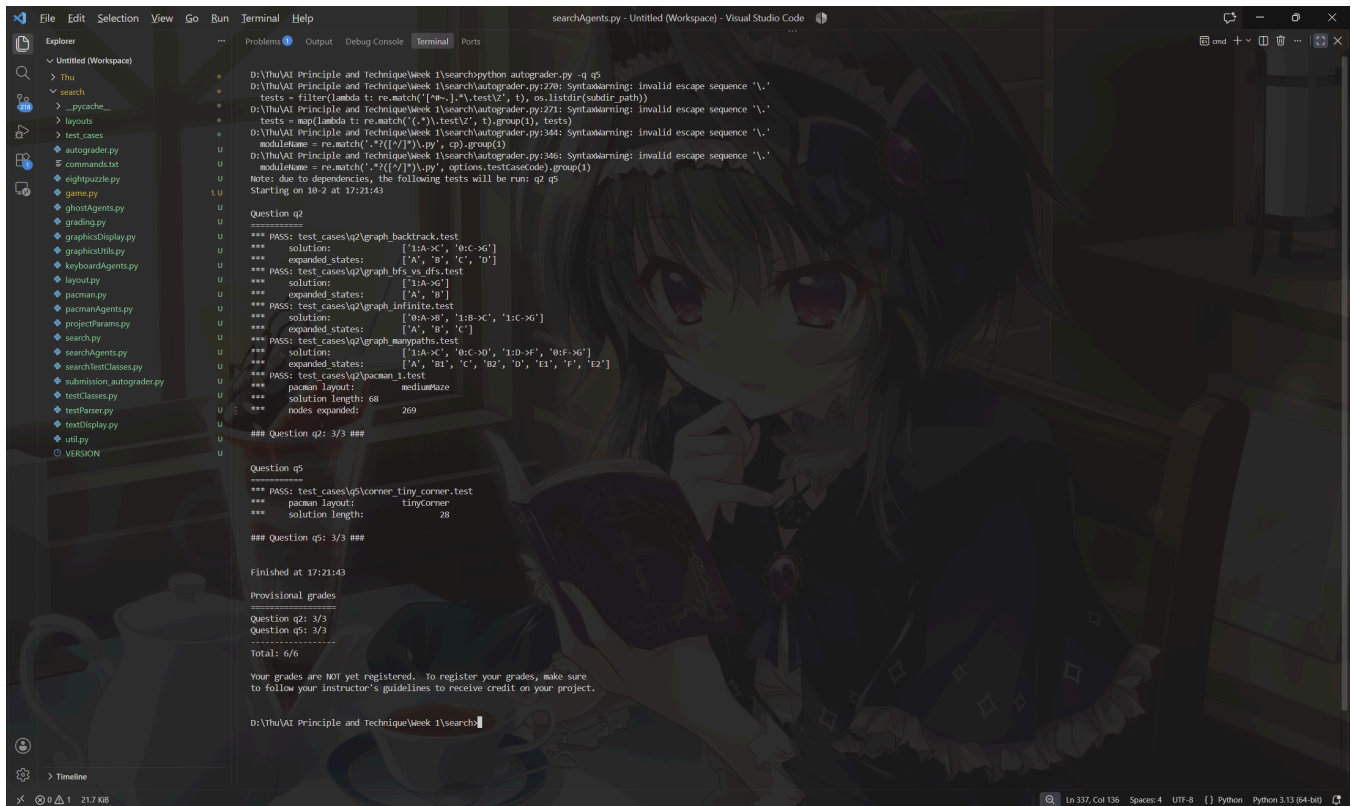
Finished at 17:00:59

Provisional grades
=====
Question q4: 3/3
-----
Total: 3/3

Your grades are NOT yet registered. To register your grades, make sure
to follow your instructor's guidelines to receive credit on your project.

D:\Thu\AI Principle and Technique\Week 1\search>
```

2.7 Question 5: Finding All the Corners



```
D:\VHU\AI Principle and Technique\Week 1\search\python autograder.py -q q5
D:\VHU\AI Principle and Technique\Week 1\search\autograder.py:268: SyntaxWarning: invalid escape sequence '\.'
  tests = filter(lambda t: re.match('^(?=[^\\test\\2, q), os.listdir(subdir_path))
D:\VHU\AI Principle and Technique\Week 1\search\autograder.py:271: SyntaxWarning: invalid escape sequence '\.'
  tests = map(lambda t: re.match('^(?=[^\\test\\2, t).group(1), tests)
D:\VHU\AI Principle and Technique\Week 1\search\autograder.py:344: SyntaxWarning: invalid escape sequence '\.'
  modulefame = re.match('^(?=[^\\test\\2, cp).group(1)
D:\VHU\AI Principle and Technique\Week 1\search\autograder.py:346: SyntaxWarning: invalid escape sequence '\.'
  modulefame = re.match('^(?=[^\\test\\2, cp), options.testcasecode).group(1)
Notes due to dependencies, the following tests will be run: q2 q5
Starting on 10-2 at 17:21:43

Question q2
====
*** PASS: test_cases\q2\graph_backtrack.test
*** solution:      ['1A-3C', '0C-3G']
*** expanded states:  ['A', 'B', 'C', 'D']
*** PASS: test_cases\q2\graph_bfs_vs_dfs.test
*** solution:      ['1A-3G']
*** expanded states:  ['A', 'B']
*** PASS: test_cases\q2\graph_infinite.test
*** solution:      ['0A-3B', '1B-3C', '1C-3G']
*** expanded states:  ['A', 'B', 'C']
*** PASS: test_cases\q2\graph_mypaths.test
*** solution:      ['1A-3G', '0C-3D', '1D-3F', '0F-3G']
*** expanded states:  ['A', 'B1', 'C', 'B2', 'D', 'E1', 'F', 'E2']
*** PASS: test_cases\q2\pacman_1.test
*** pacman layout:  mediummaze
*** solution length: 68
*** nodes expanded: 269

## Question q2: 3/3 ##

Question q5
====
*** PASS: test_cases\q5\corner_tiny_corner.test
*** pacman layout:  tinycorner
*** solution length: 28

## Question q5: 3/3 ##

Finished at 17:21:43

Provisional grades
====
Question q2: 3/3
Question q5: 3/3
-----
Total: 6/6

Your grades are NOT yet registered. To register your grades, make sure
to follow your instructor's guidelines to receive credit on your project.

D:\VHU\AI Principle and Technique\Week 1\search\
```

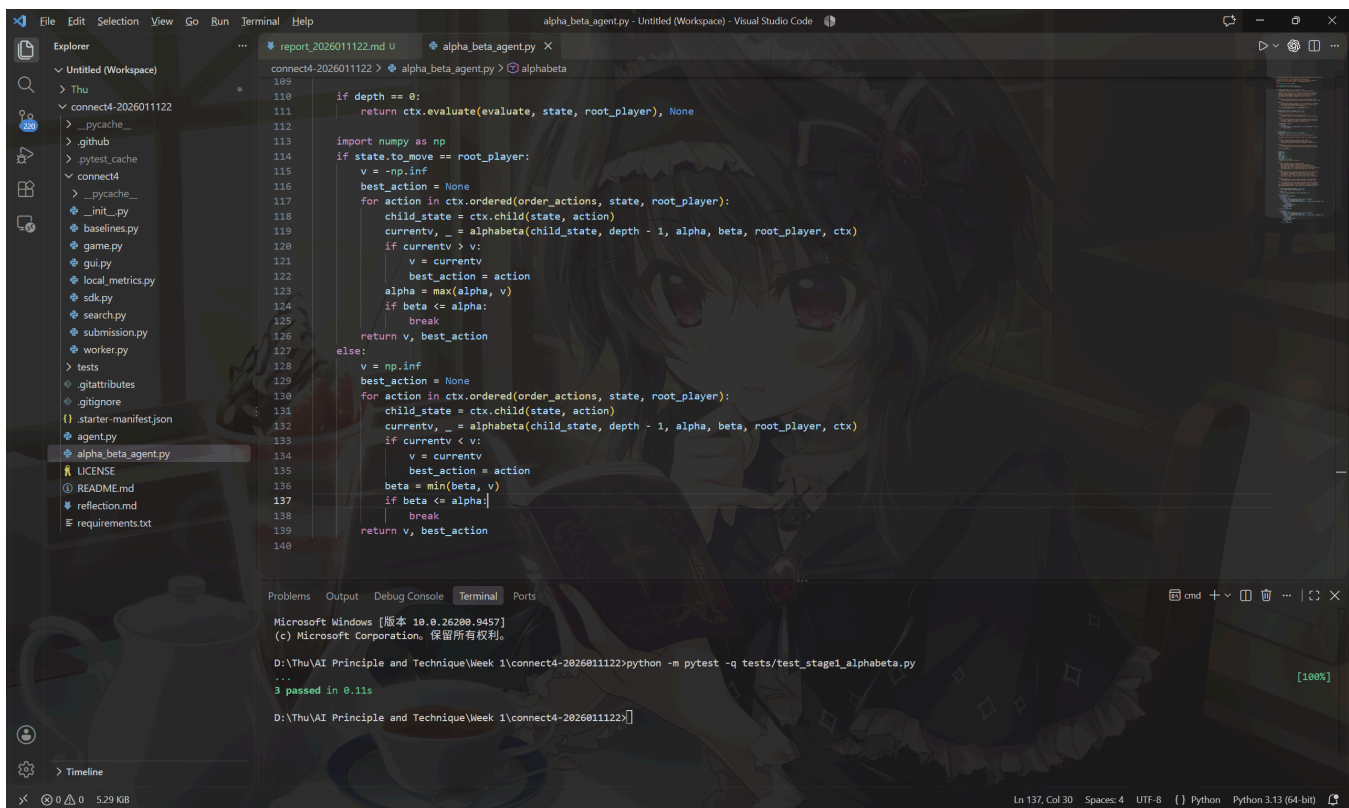
2.8 Question 6: Corners Problem: Heuristic

The image shows a Windows desktop environment. In the foreground, a VS Code editor window is open, displaying a Python script named 'autograder.py'. The script contains several syntax warnings and test results. The background features a large, stylized anime-style illustration of a girl with long, wavy purple hair, wearing a white and blue dress with a large white collar and a blue sash. She has a gentle expression and is holding a small white object. The desktop taskbar at the bottom shows the Start button, a search bar, and several pinned applications including File Explorer, Edge, and VS Code. The system tray on the right shows the date and time as 11:51 AM on 10/2/2023.

3. Programming Component 2: Connect Four

3.1 Play a Few Games First

3.2 Question 1: Alpha-Beta Search



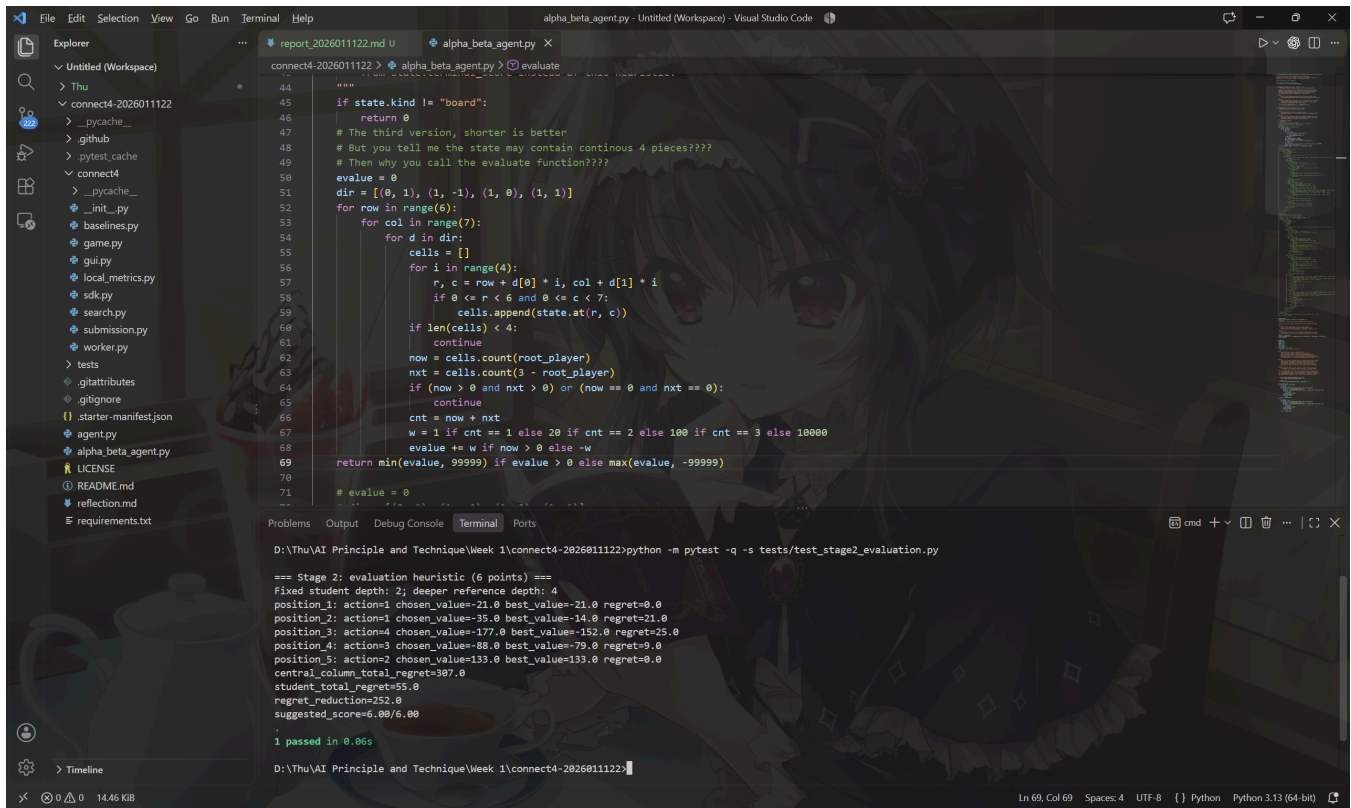
```
connect4-2026011122 > alpha_beta_agent.py > alphabeta
109
110 if depth == 0:
111     return ctx.evaluate(evaluate, state, root_player), None
112
113 import numpy as np
114 if state.to_move == root_player:
115     v = -np.inf
116     best_action = None
117     for action in ctx.ordered(order_actions, state, root_player):
118         child_state = ctx.child(state, action)
119         currentv, _ = alphabeta(child_state, depth - 1, alpha, beta, root_player, ctx)
120         if currentv > v:
121             v = currentv
122             best_action = action
123             alpha = max(alpha, v)
124             if beta <= alpha:
125                 break
126         return v, best_action
127 else:
128     v = np.inf
129     best_action = None
130     for action in ctx.ordered(order_actions, state, root_player):
131         child_state = ctx.child(state, action)
132         currentv, _ = alphabeta(child_state, depth - 1, alpha, beta, root_player, ctx)
133         if currentv < v:
134             v = currentv
135             best_action = action
136             beta = min(beta, v)
137             if beta <= alpha:
138                 break
139     return v, best_action
140
```

Microsoft Windows [版本 10.0.26200.9457]
(c) Microsoft Corporation. 保留所有权利。

D:\Thy\AI Principle and Technique\Week 1\connect4-2026011122>python -m pytest -q tests/test_stage1_alphabeta.py
...
3 passed in 0.11s

D:\Thy\AI Principle and Technique\Week 1\connect4-2026011122>

3.3 Question 2: Improve Evaluation



```
alpha_beta_agent.py - Untitled (Workspace) - Visual Studio Code
connect4-2026011122 > alpha_beta_agent.py > evaluate
44  """
45  if state.kind != "board":
46      return 0
47  # The third version, shorter is better
48  # But you tell me the state may contain continuous 4 pieces???
49  # Then why you call the evaluate function???
50  evalue = 0
51  dir = [(0, 1), (1, -1), (1, 0), (1, 1)]
52  for row in range(6):
53      for col in range(7):
54          for d in dir:
55              cells = []
56              for i in range(4):
57                  r, c = row + d[0] + i, col + d[1] + i
58                  if 0 <= r < 6 and 0 <= c < 7:
59                      cells.append(state.at(r, c))
60                  if len(cells) < 4:
61                      continue
62                  now = cells.count(root_player)
63                  nxt = cells.count(3 - root_player)
64                  if (now > 0 and nxt > 0) or (now == 0 and nxt == 0):
65                      continue
66                  cnt = now + nxt
67                  w = 1 if cnt == 1 else 20 if cnt == 2 else 100 if cnt == 3 else 10000
68                  evalue += w if now > 0 else -w
69  return min(evalue, 99999) if evalue > 0 else max(evalue, -99999)
70
71  # evalue = 0
72  """

D:\Th\AI Principle and Technique\Week 1\connect4-2026011122>python -m pytest -q -s tests/test_stage2_evaluation.py

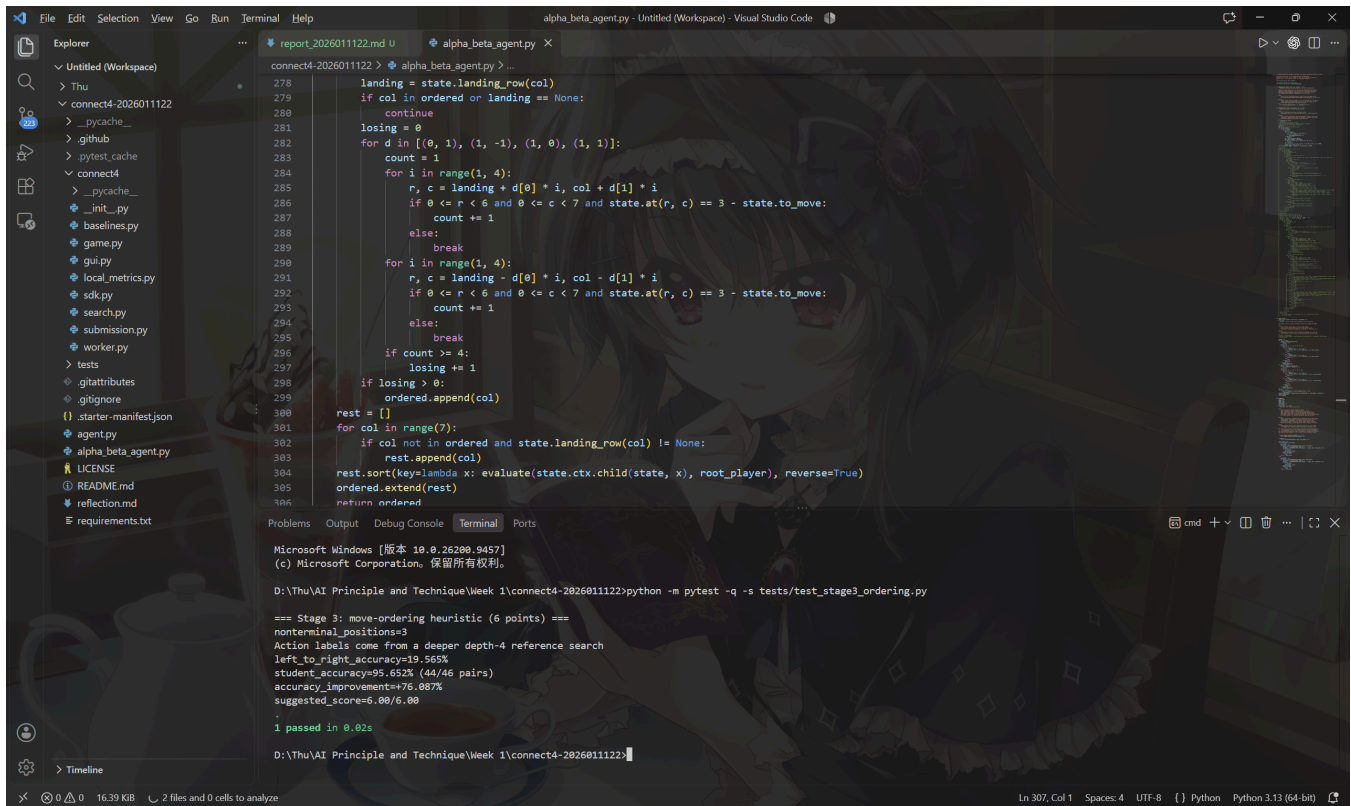
=== Stage 2: evaluation heuristic (6 points) ===
Fixed student depth: 2; deeper reference depth: 4
position_1: action=1 chosen_value=-21.0 best_value=-21.0 regret=0.0
position_2: action=1 chosen_value=-35.0 best_value=-14.0 regret=21.0
position_3: action=4 chosen_value=-177.0 best_value=-152.0 regret=25.0
position_4: action=3 chosen_value=-88.0 best_value=-79.0 regret=9.0
position_5: action=2 chosen_value=133.0 best_value=133.0 regret=0.0
central_column_total_regret=307.0
student_total_regret=55.0
regret_reduction=252.0
suggested_score=6.00/6.00
.
1 passed in 0.06s

D:\Th\AI Principle and Technique\Week 1\connect4-2026011122>
```

My evaluation is that for every potentially becoming connected-4 component (that is the continuous 4-grids only contain one's piece and space), if there is 1 pieces, count 1 score; 2 pieces, count 20 score; 3 pieces, count 100 score; 4 pieces, count 10000 score.

This method can handle the difference between live 2 and dead 2 (a.k.a. 活二 and 冲二) and so on, so it is a reasonable evaluation.

3.4 Question 3: Improve Move Ordering



```
connect4_2026011122 > alpha_beta_agent.py ...
278 landing = state.landing_row(col)
279 if col in ordered or landing == None:
280     continue
281 losing = 0
282 for d in [(0, 1), (1, -1), (1, 0), (1, 1)]:
283     count = 1
284     for i in range(1, 4):
285         r, c = landing + d[0] * i, col + d[1] * i
286         if 0 <= r < 6 and 0 <= c < 7 and state.at(r, c) == 3 - state.to_move:
287             count += 1
288         else:
289             break
290     for i in range(1, 4):
291         r, c = landing - d[0] * i, col - d[1] * i
292         if 0 <= r < 6 and 0 <= c < 7 and state.at(r, c) == 3 - state.to_move:
293             count += 1
294         else:
295             break
296     if count >= 4:
297         losing += 1
298     if losing > 0:
299         ordered.append(col)
300 rest = []
301 for col in range(7):
302     if col not in ordered and state.landing_row(col) != None:
303         rest.append(col)
304 rest.sort(key=lambda x: evaluate(state.ctx.child(state, x), root_player), reverse=True)
305 ordered.extend(rest)
306 return ordered

Problems Output Debug Console Terminal Ports
Microsoft Windows [版本 10.0.26200.9457]
(c) Microsoft Corporation. 保留所有权利。

D:\Thu\AI Principle and Technique\Week 1\connect4-2026011122>python -m pytest -q -s tests/test_stage3_ordering.py

=== Stage 3: move-ordering heuristic (6 points) ===
nonterminal_positions=3
Action labels come from a deeper depth-4 reference search
left_to_right_accuracy=95.65%
student_accuracy=95.652% (44/46 pairs)
accuracy_improvement=76.987%
suggested_score=6.00/6.00

1 passed in 0.02s

D:\Thu\AI Principle and Technique\Week 1\connect4-2026011122>
```

For a action that the player will win, or a action that the player must to take to avoid losing, they are at the beginning of the order. For the rest, I simply sort all actions with the evaluation function. Although to do this will lose nearly half of states count, it did well on scoring, getting 95% accuracy. And since the pairwise accuracy is high, it can prune a lot states, so the trade-off is worthy.